

# Experiment Report

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**Course:** Big Data Analysis Technology

**Experiment Topic:** Social Network Analysis

## Questions

1. What is Social Network Analysis? For a single node, what can we analyze in Social Network? For the structure of social Network, what can we analyze for Social Network? Explain all the metrics with examples.

Social Network Analysis (SNA) is a discipline that studies social structure and relationships by examining patterns of connections between individuals.

### Node-Level Analysis:

When examining a single individual within a social network, we can use various metrics to understand their role and influence:

- **Degree:** This simple metric quantifies the number of connections an individual has, providing a basic measure of their social activity.  
For example, a student with a high degree might be considered popular or well-connected in their school.
- **Centrality Measures:** Centrality metrics go beyond simple connections to assess an individual's importance within the network.
- **Degree Centrality:** This measure simply counts the number of direct connections a node has.
- **Betweenness Centrality:** High betweenness centrality suggests a node is a "bridge" or connector, often acting as a conduit for information flow.  
For instance, a student with high betweenness centrality might be the central person in a gossip network.
- **Closeness Centrality:** Individuals with low closeness centrality are centrally located in the network, enabling them to reach other nodes quickly. They may be well-informed or have a broad range of contacts.
- **Eigenvector Centrality:** This measure reflects a node's influence based on its connections to other influential nodes.  
A person with high Eigenvector centrality might be considered a thought leader or an influencer.

- **Clustering Coefficient:** This metric tells us how interconnected an individual's immediate network is. A high clustering coefficient suggests they are part of a tight-knit group.
- **Tie Strength:** This metric assesses the strength or importance of the connection between two individuals. Stronger ties might involve frequent interaction, emotional closeness, or shared experiences.

### Network Structure Analysis:

Examining the overall structure of a social network reveals insights into its organization and dynamics:

- **Density:** Density measures how many connections exist compared to the maximum possible connections.  
High density signifies a closely knit network, like a small, tight-knit community where everyone knows everyone else.
- **Diameter:** This metric represents the longest shortest path between any two nodes in the network. A small diameter indicates that information can travel quickly throughout the network.
- **Average Path Length:** This metric calculates the average distance between all pairs of nodes. A low average path length suggests that the network is well-connected and information can easily spread.
- **Modularity:** Modularity quantifies how well the network is divided into distinct communities.  
High modularity indicates that the network is organized into clearly identifiable groups, with strong connections within each group and weaker connections between them.
- **Network Motifs:** Recurring patterns of connections, known as network motifs, reveal common social interactions or structures within the network. For example, a “triangle” motif, where three nodes are connected to each other, might represent a strong social bond or a shared interest.

### Example:

Imagine analyzing the social network of a university campus. Using SNA, we could:

- **Analyze individual students:** Identify influential students (high Eigenvector centrality), students who act as information bridges (high betweenness centrality), or students part of close-knit friend groups (high clustering coefficient).

- **Analyze network structure:** Determine if the network is densely connected, identify student clubs and organizations as distinct communities, or detect emerging trends in student social interaction.

2. Compared Social Network with other Network, what special characteristics does the Social Network have?

**Network:** Composition of several nodes and links connecting these nodes.

**Social network:** A network system formed among individual members of society through social relationships.

### **Special characteristics of Social Network:**

- **Relationships:** Social networks are built on relationships between people. The connections represent interactions, friendships, family ties, professional affiliations, etc.
- **Dynamic Nature:** Social networks are constantly evolving. People form new connections, break old ones, and their relationships change over time. This adds a level of dynamism that isn't always present in other networks.
- **Intentionality:** Social networks are often formed intentionally. People choose who they connect with, and those connections often have a purpose (e.g., friendship, collaboration, support).
- **Multifaceted Interactions:** Social networks involve diverse interactions. People communicate, share information, collaborate, influence each other, and engage in various forms of social behavior. This creates a richness of information and a complex web of relationships.

While all networks consist of nodes and connections, social networks exhibit distinct characteristics. They are built on social relationships between individuals, emphasizing the intentionality and emotional depth of those connections. Unlike networks like transportation or communication systems, social networks are **dynamic**, evolving as relationships change. They are also characterized by **multifaceted interactions, encompassing communication, information sharing, collaboration, and influence.**

# Experiment 1

**Experiment topic:** Community Detection with NetworkX in Python.

I did the experiment in PyCharm. Installing networkX and Matplotlib.

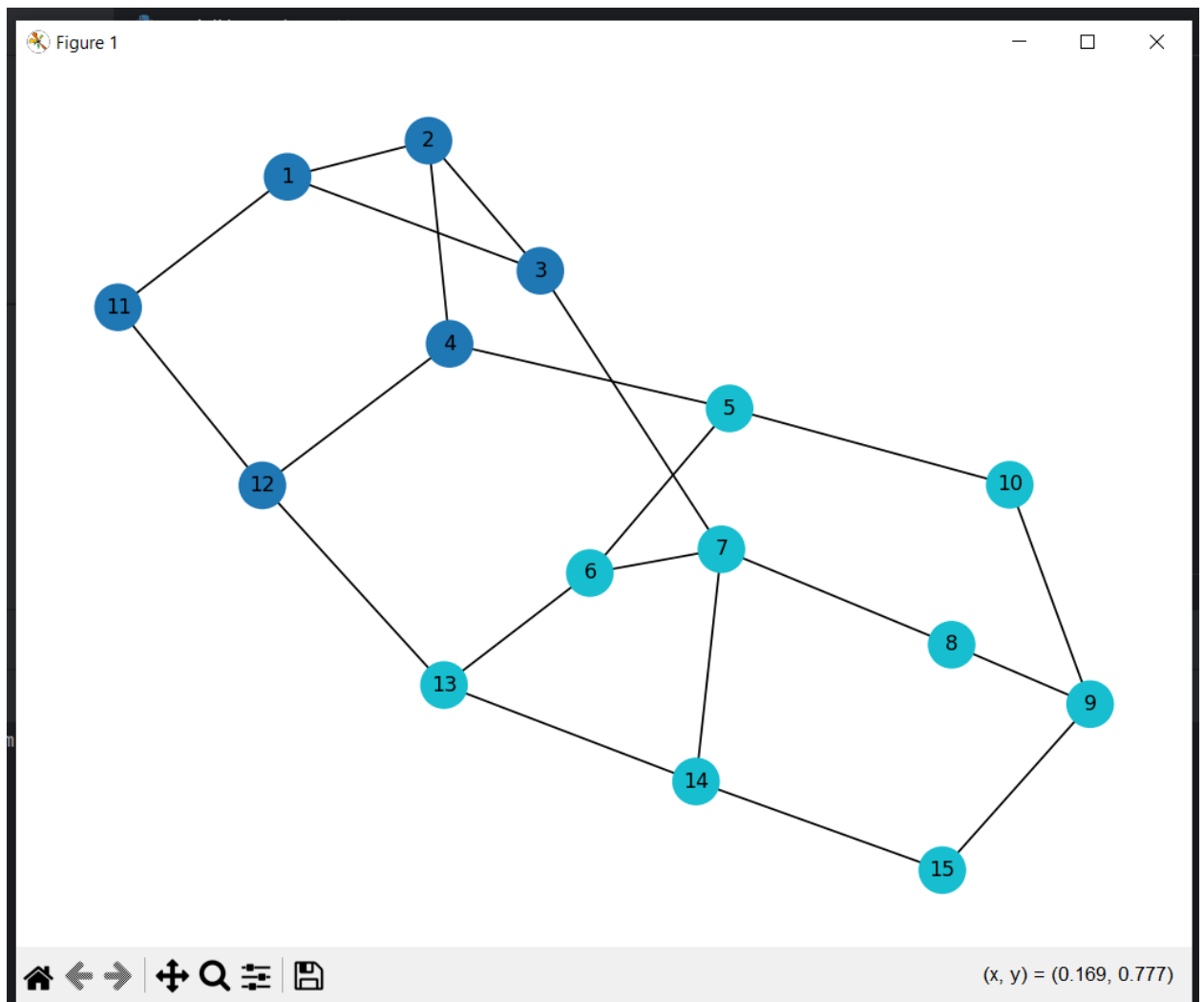
```
(venv) PS C:\Users\prett\PycharmProjects\rucharm> pip install --upgrade networkx matplotlib
Requirement already satisfied: networkx in c:\users\prett\pycharmprojects\rucharm\venv\lib\site-packages (3.4.1)
Collecting networkx
  Downloading networkx-3.4.2-py3-none-any.whl.metadata (6.3 kB)
Requirement already satisfied: matplotlib in c:\users\prett\pycharmprojects\rucharm\venv\lib\site-packages (3.9.0)
Collecting matplotlib
```

Successfully installed matplotlib-3.9.2 networkx-3.4.2

Running the given code.

```
socialNetwork.py ×
1 import networkx as nx
2 import matplotlib.pyplot as plt
3 from networkx.algorithms import community
4
5 # Create graph G
6 G = nx.Graph() # Undirected graph without multiple edges
7 G.add_node(1)
8 G.add_nodes_from([2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15])
9 G.add_edges_from([(1, 2), (4, 5), (5, 6), (2, 4), (1, 3), (2, 3),
10                  (6, 7), (7, 8), (8, 9), (9, 10), (5, 10), (3, 7),
11                  (1, 11), (11, 12), (12, 4), (12, 13), (13, 6),
12                  (13, 14), (14, 7), (14, 15), (9, 15)])
13
14 # Use Girvan-Newman algorithm to divide communities
15 communities_generator = community.girvan_newman(G)
16 top_level_communities = next(communities_generator)
17 community_mapping = {node: i for i, community in enumerate(top_level_communities) for node in community}
18
19 # Sets the color of the node, based on the community division
20 node_colors = [community_mapping[node] for node in G.nodes]
21
22 # Visualize the network and mark communities with different colors
23 plt.figure(figsize=(8, 6))
24 pos = nx.spring_layout(G) # Plot using the spring layout
25 nx.draw(G, pos, with_labels=True, node_color=node_colors, cmap=plt.cm.tab10, node_size=500, font_size=10)
26 plt.title("Community Detection in Modified Graph")
27 plt.show()
28
29 # Print the community division results
30 print("The results of community segmentation: ")
```

Resulting output:



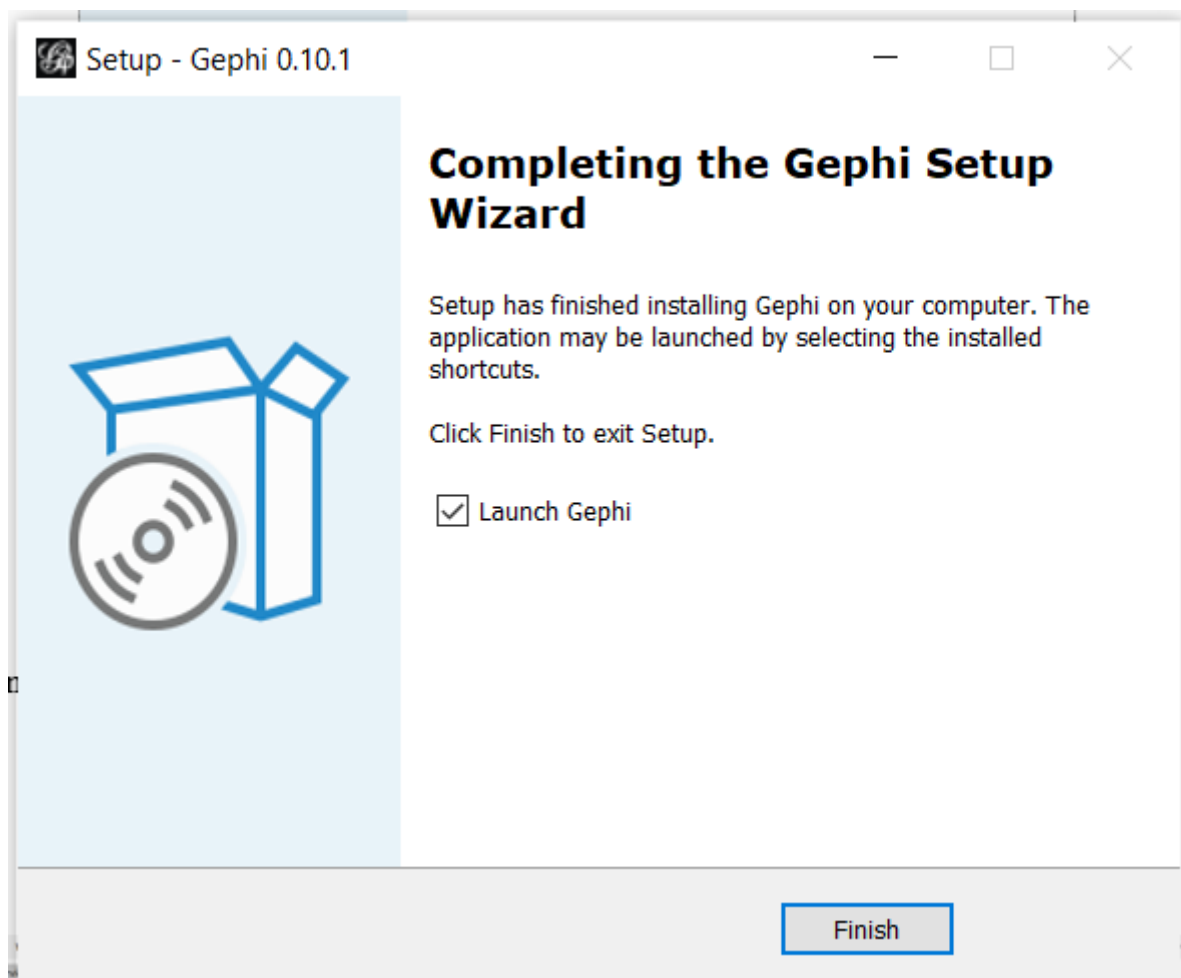
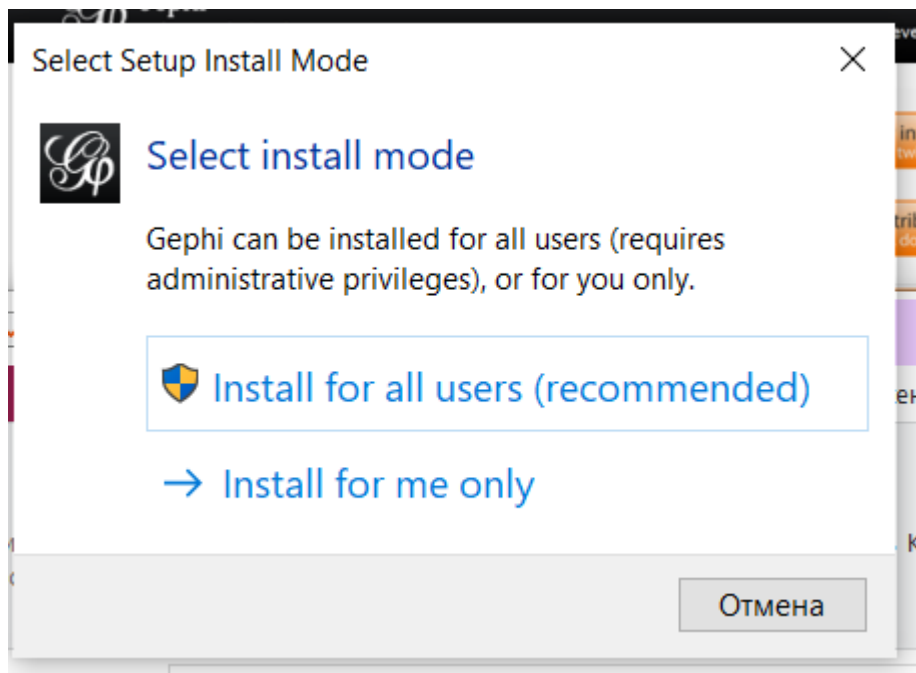
```
Run socialNetwork x
C:\Users\prett\PycharmProjects\rucharm\venv\Scripts\python.exe C:\Users\prett\PycharmProjects\rucharm\socialNetwork.py
The results of community segmentation:
Community 1: [1, 2, 3, 4, 11, 12]
Community 2: [5, 6, 7, 8, 9, 10, 13, 14, 15]
Process finished with exit code 0
```

The generated plot showed the graph with nodes colored according to the community division. Nodes within the same community have the same color, while nodes in different communities have distinct colors.

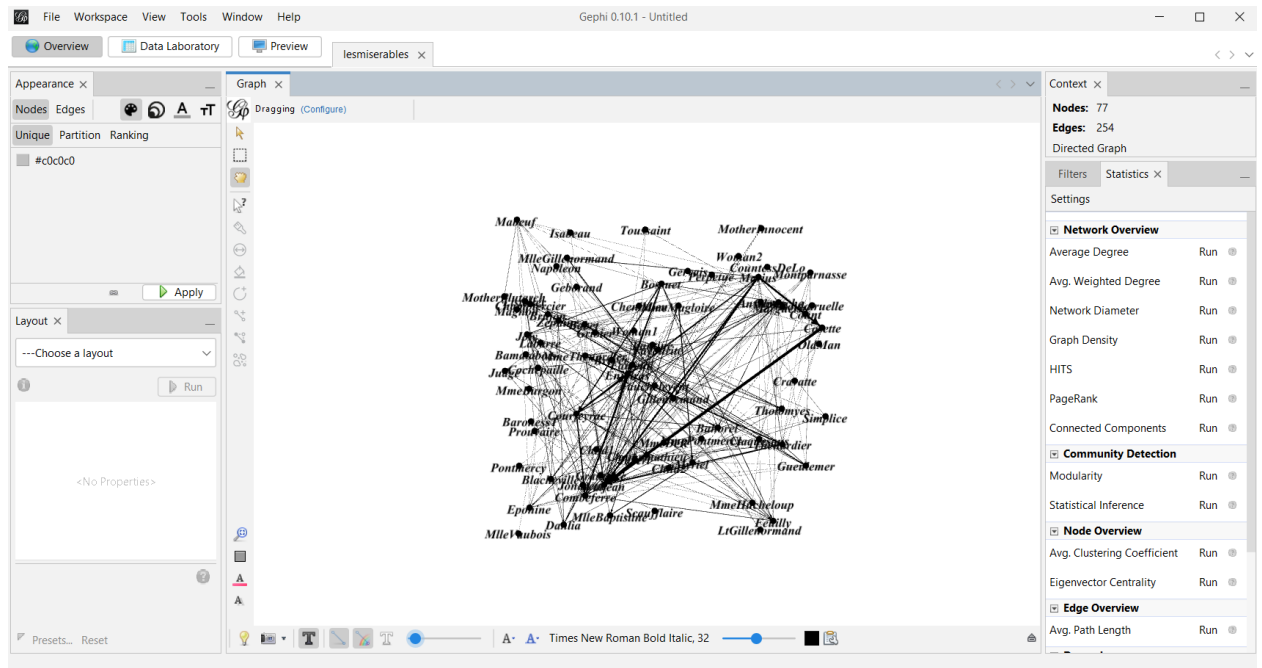
## Experiment 2

**Experiment topic:** SNA Graph Analysis Using Gephi

Installing Gephi.

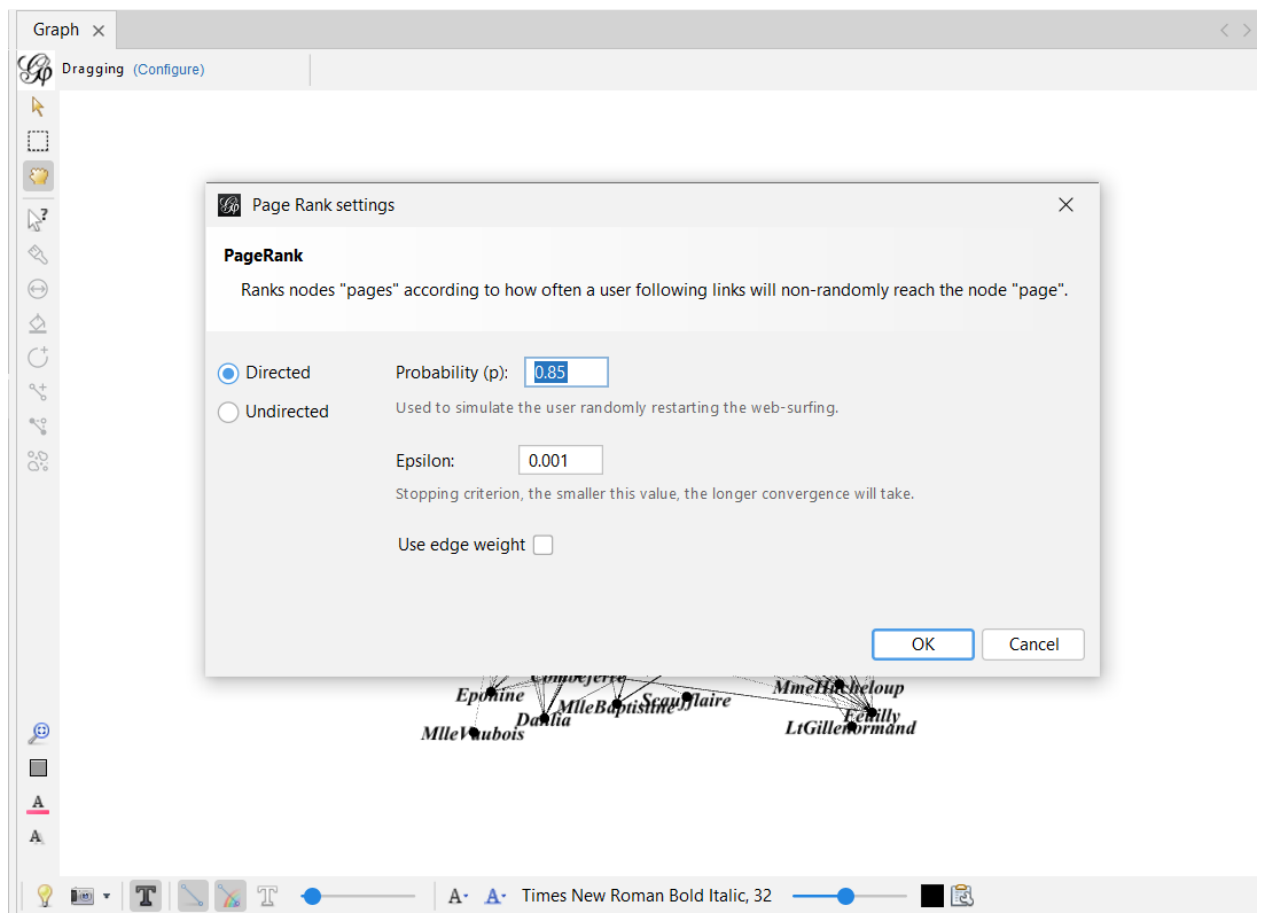


Opening graph file “*lesmiserables.gml*” and adding labels.



## 1. PageRank analysis.

Running PageRank.

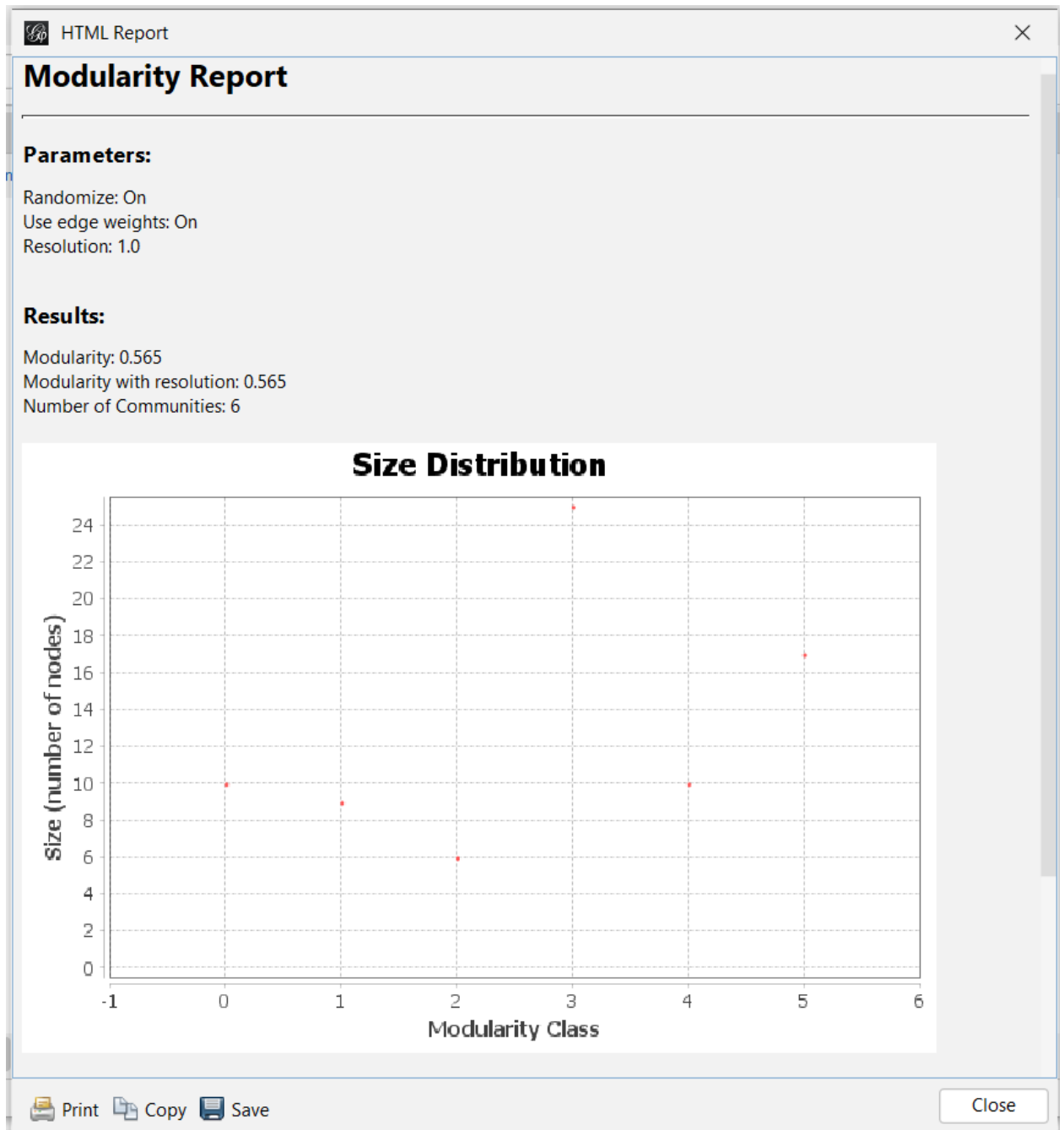


Results:





Result:



The analysis indicates a network structure with 6 communities of different sizes, with a moderate level of modularity (0.565). So, the network is not completely random.

3. Average Degree analysis

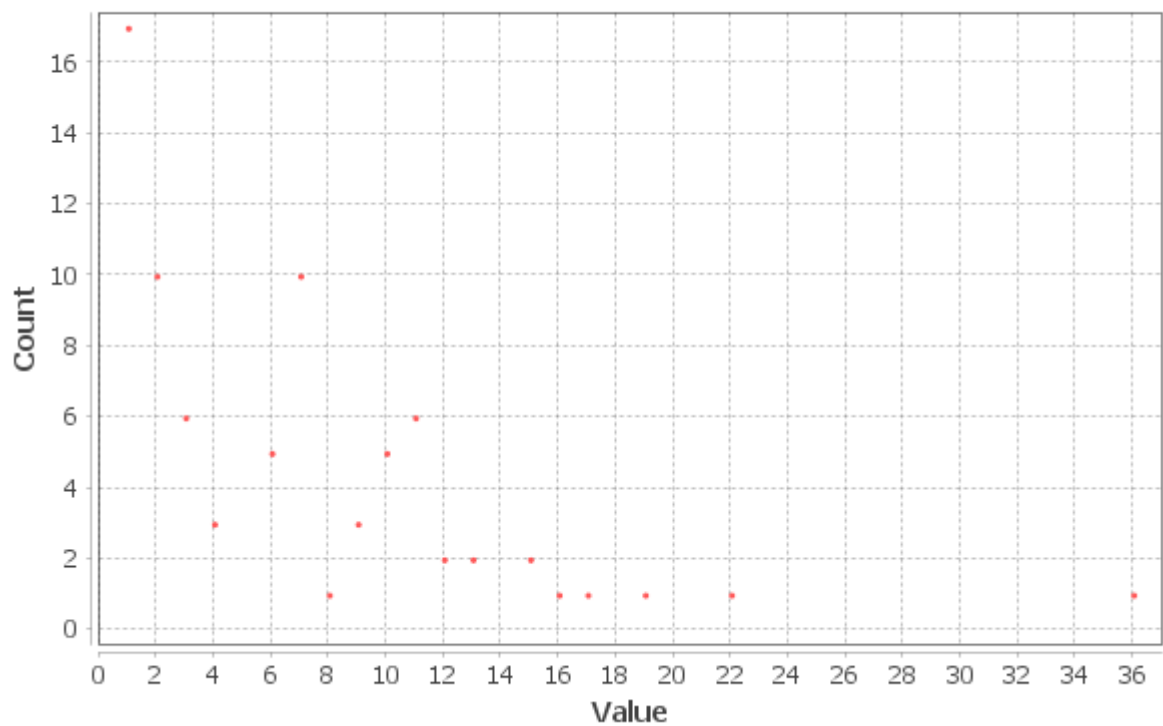
 HTML Report

# Degree Report

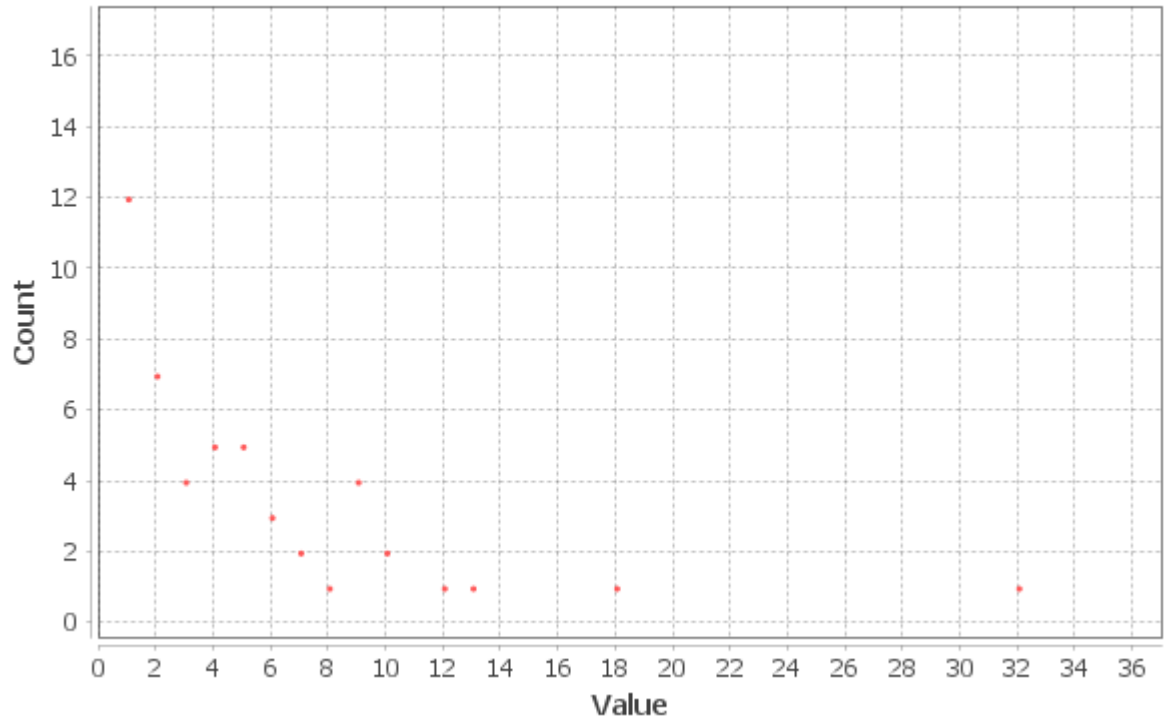
**Results:**

Average Degree: 3.299

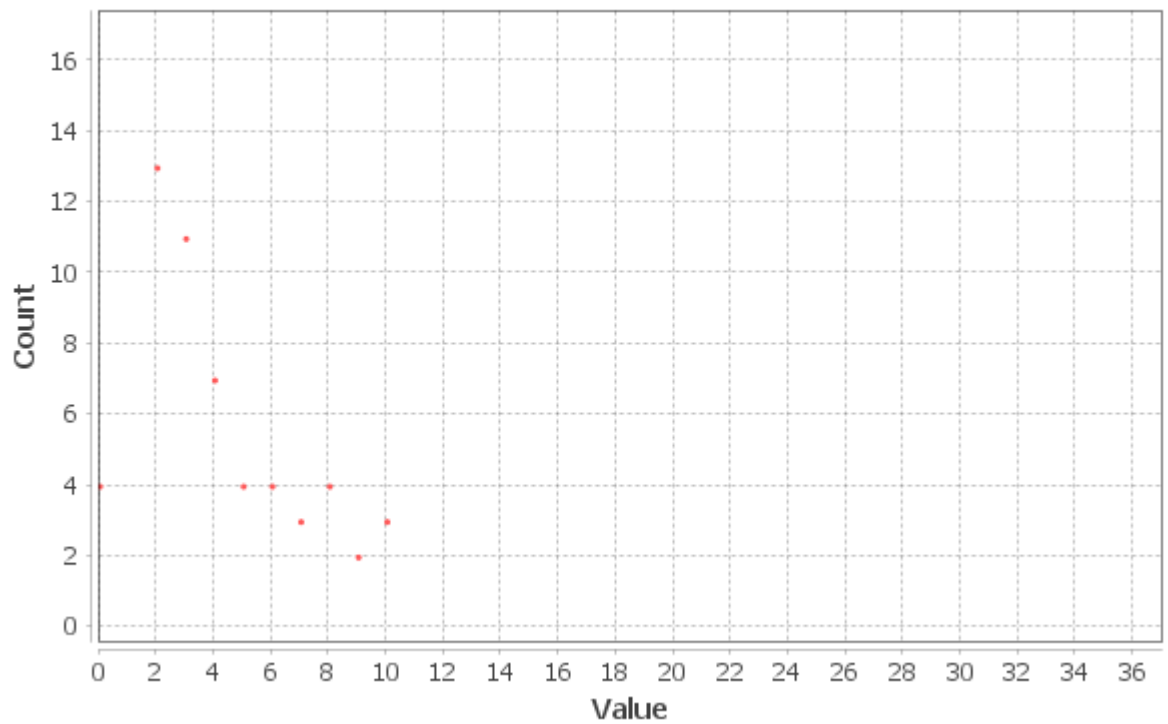
Degree Distribution



## In-Degree Distribution



## Out-Degree Distribution



An average degree of 3.299 suggests a moderately connected network.

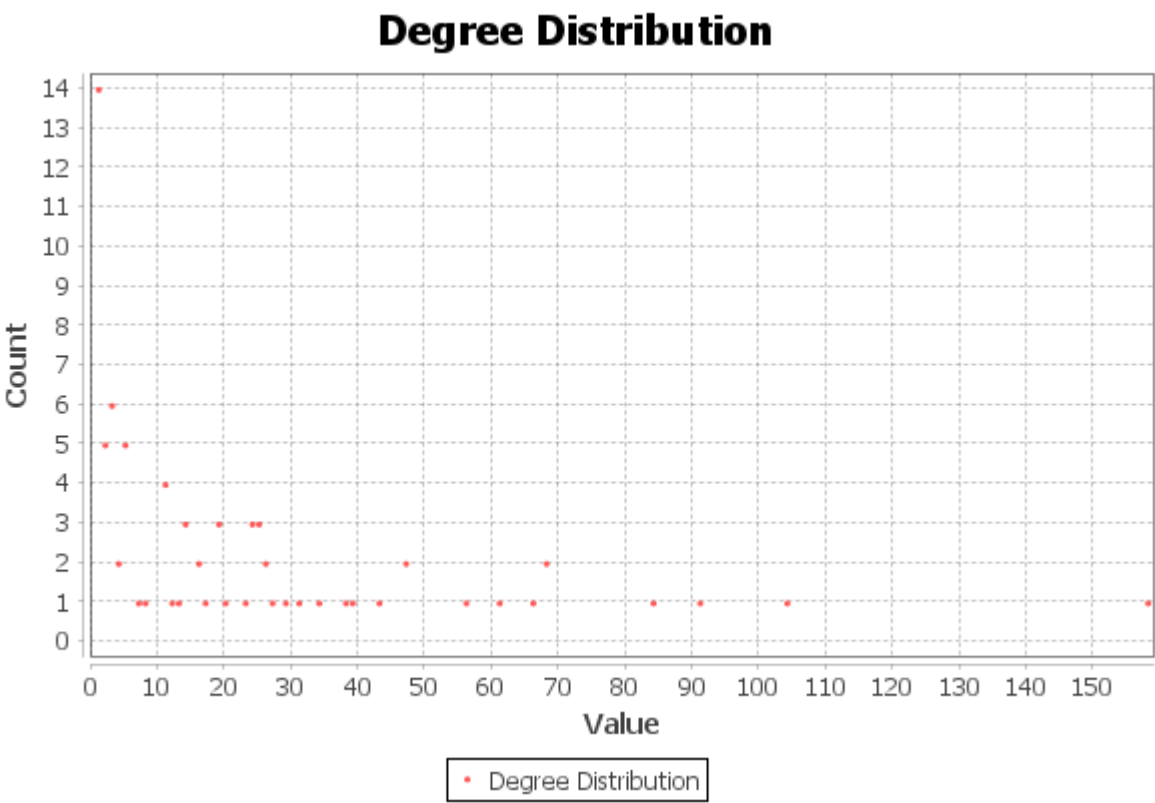
4. Average Weighted Degree analysis

 HTML Report

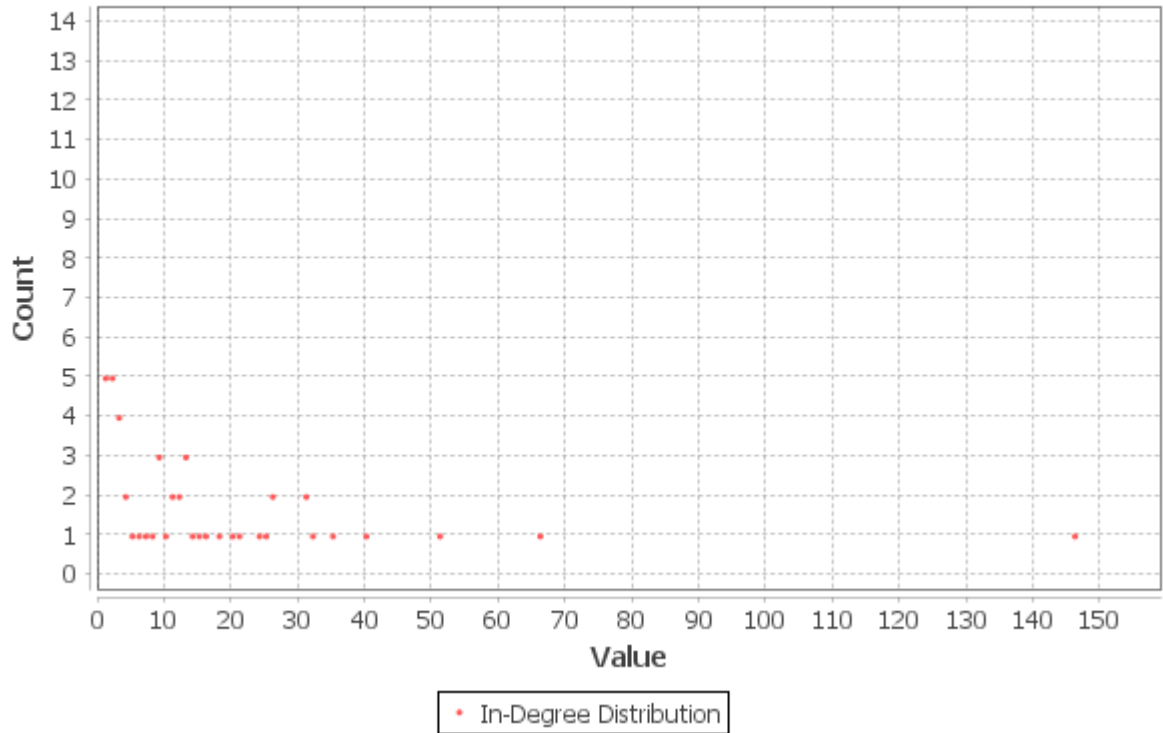
# Weighted Degree Report

**Results:**

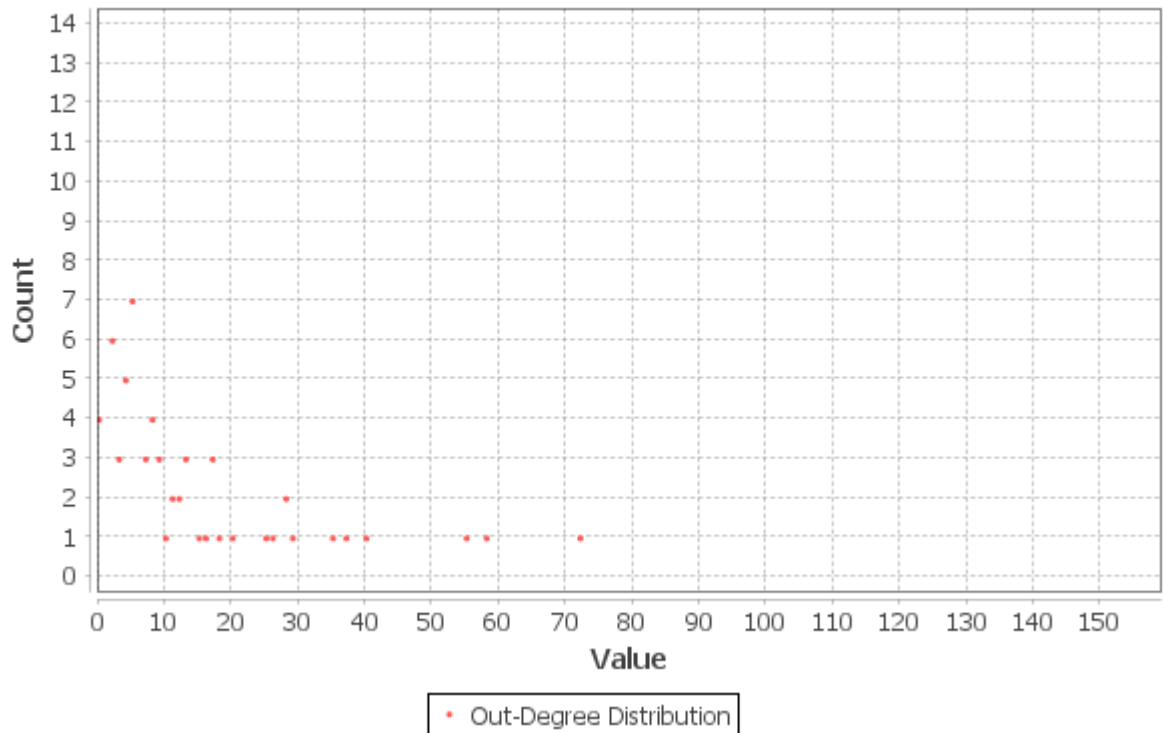
Average Weighted Degree: 10.649



## In-Degree Distribution



## Out-Degree Distribution



An average weighted degree of 10.649 suggests a network with strong connections between its nodes.

5. Network Diameter analysis

Go HTML Report

Graph Distance Report

Parameters:

Network Interpretation: directed

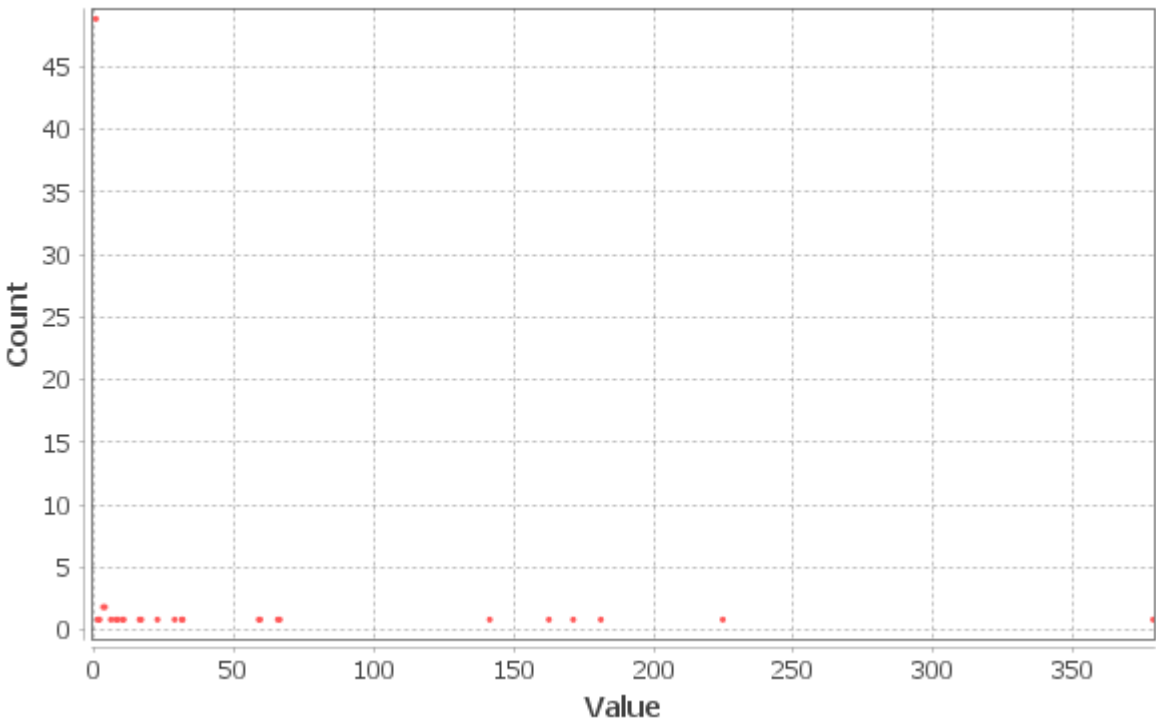
Results:

Diameter: 5

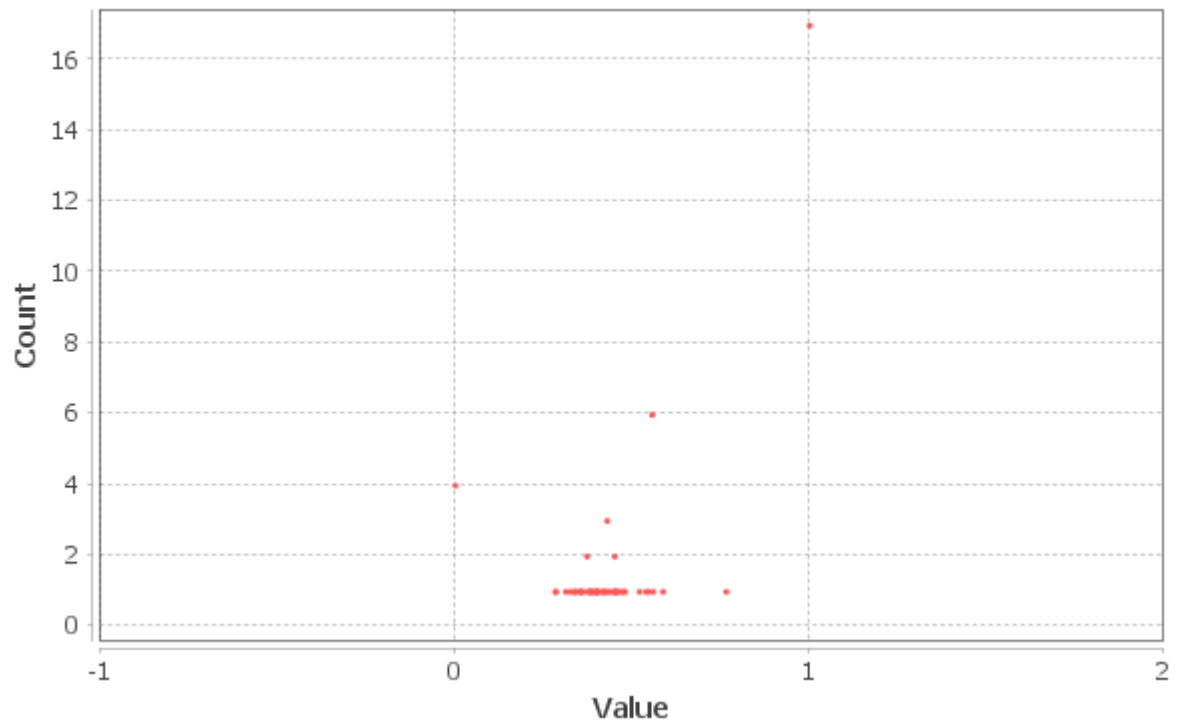
Radius: 0

Average Path length: 2.4

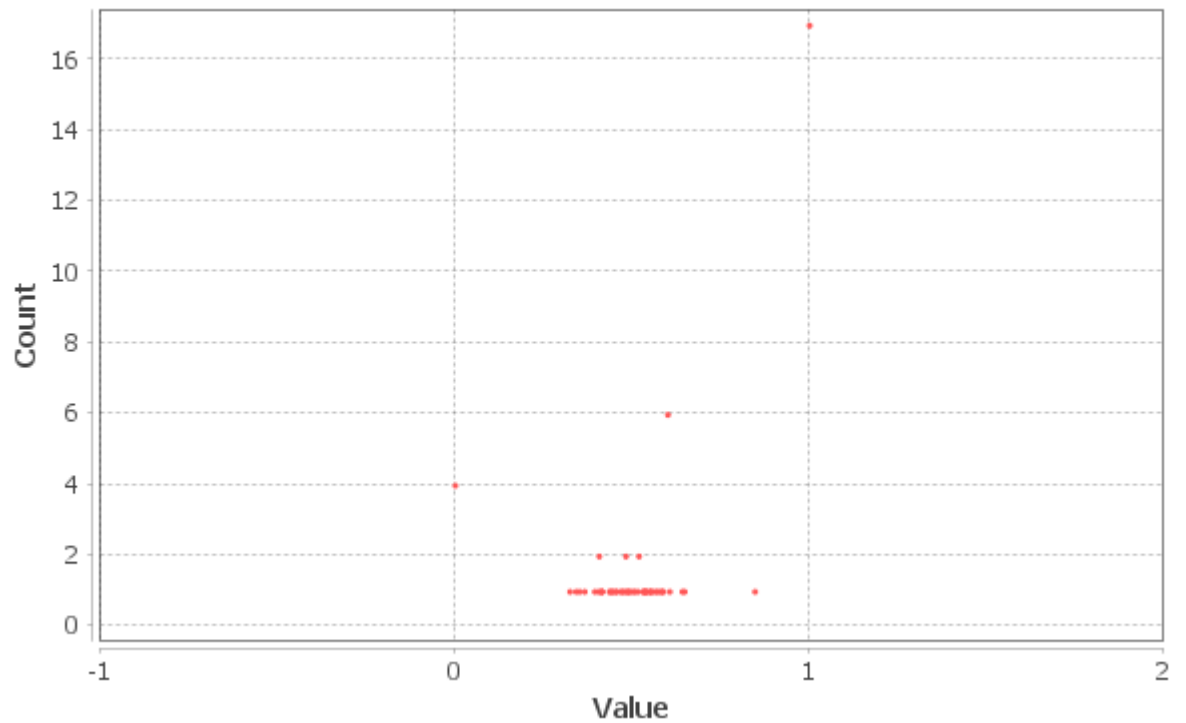
Betweenness Centrality Distribution

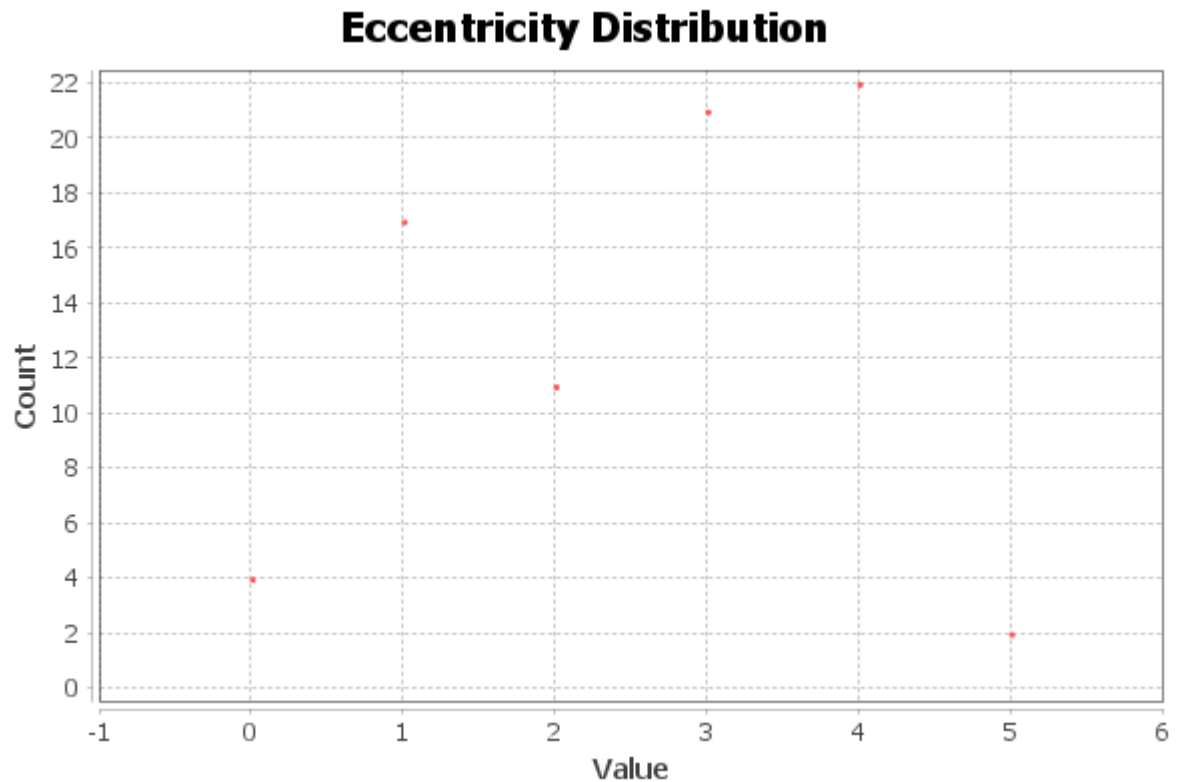


**Closeness Centrality Distribution**



**Harmonic Closeness Centrality Distribution**



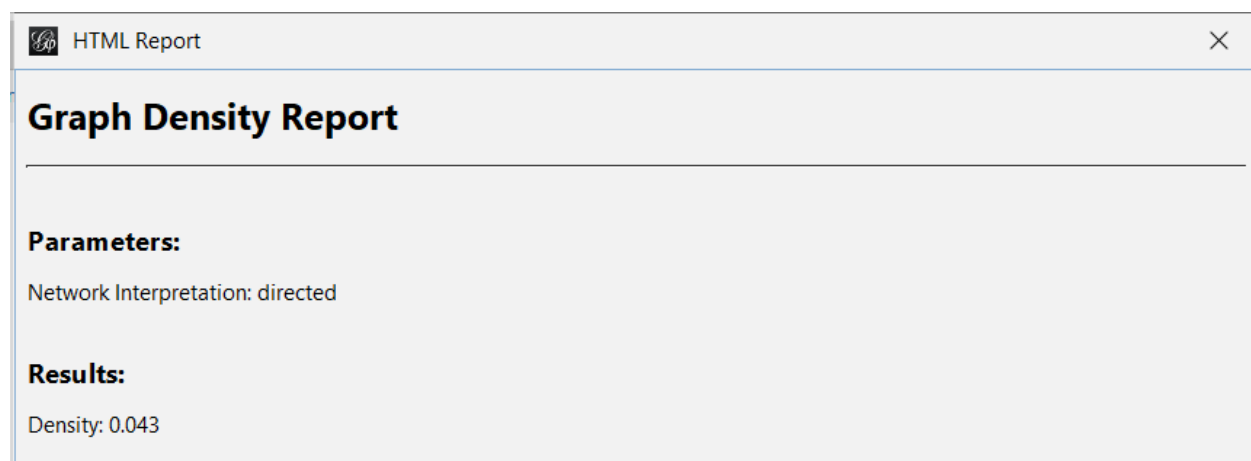


The diameter is the longest shortest path in the network. A diameter of 5 suggests that the network is relatively small and well-connected.

The radius of 0 indicates that the network is centered around a single highly connected node.

The 2.4 average path length suggests that nodes are, on average, close to each other.

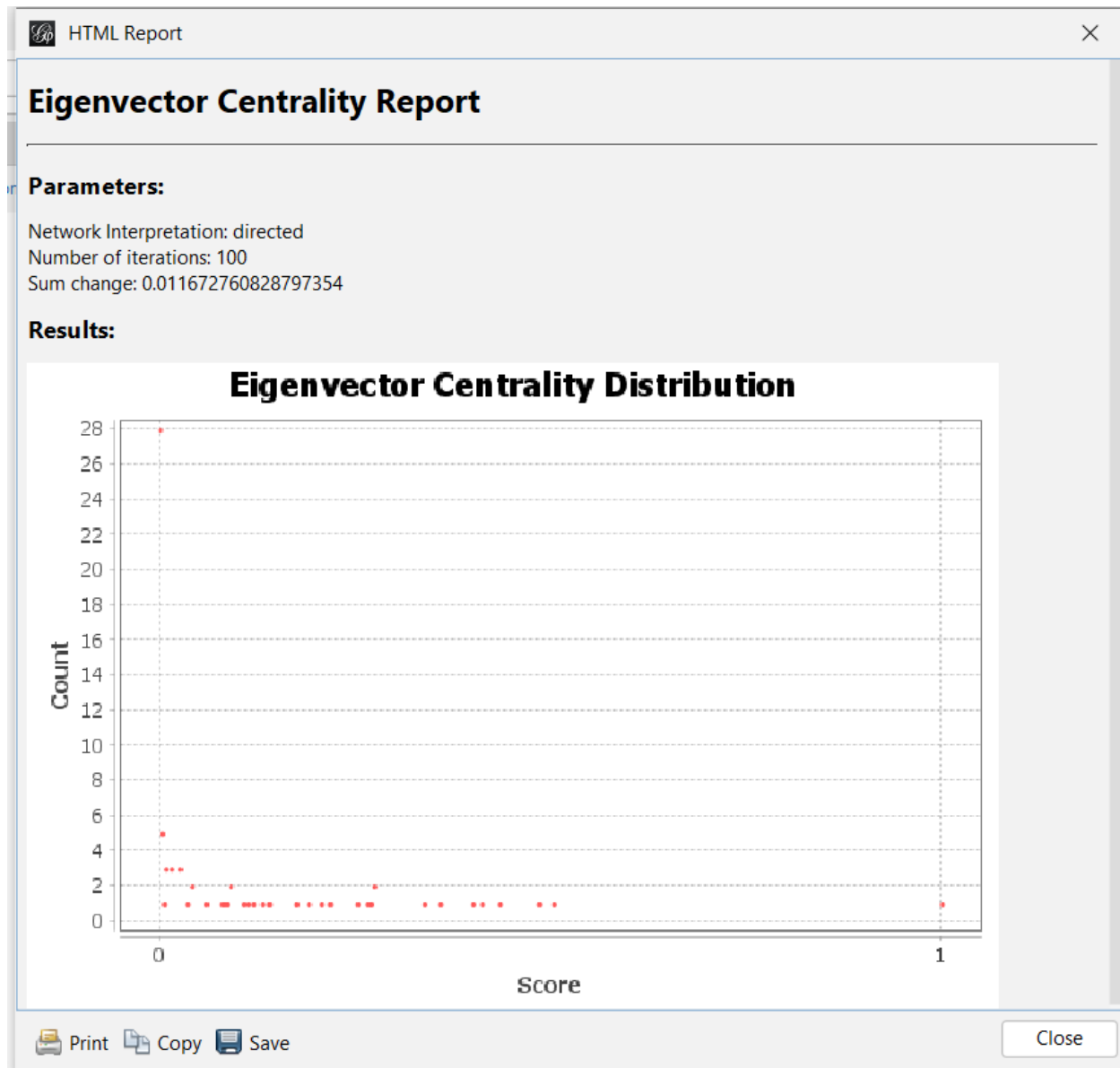
## 6. Graph Density analysis



The network has a relatively small number of connections compared to the maximum possible. The density of 0.043 means that the nodes are not very densely interconnected.



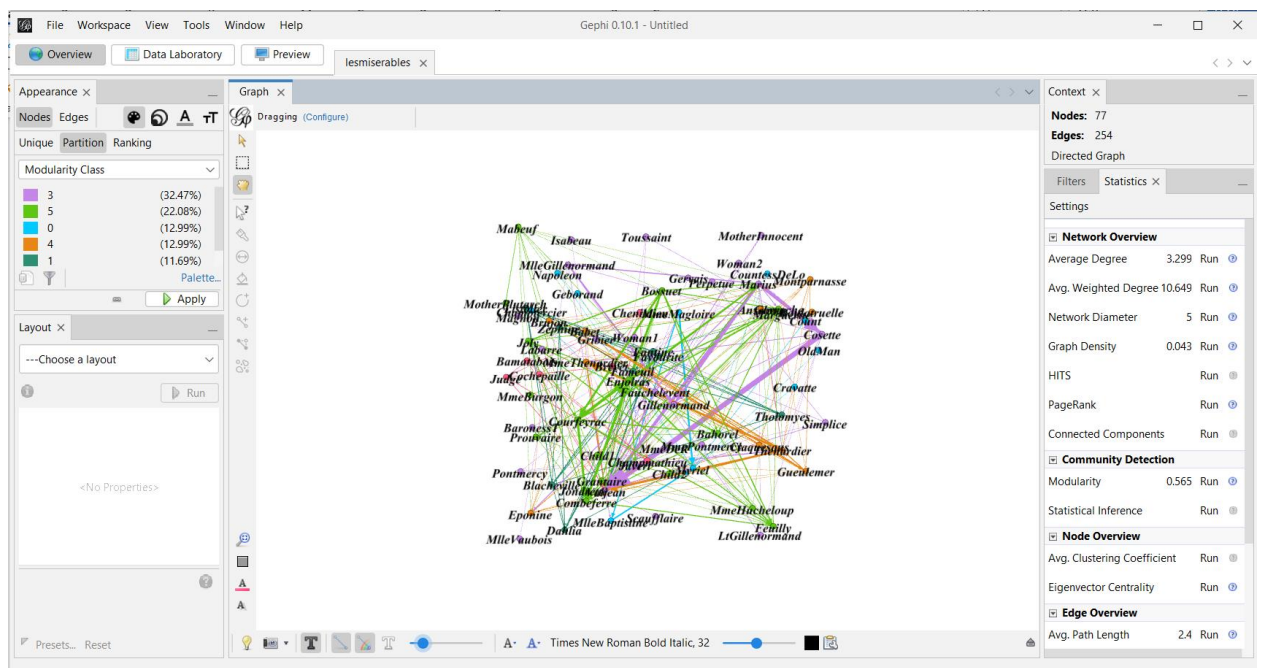
## 7. Eigenvector Centrality analysis



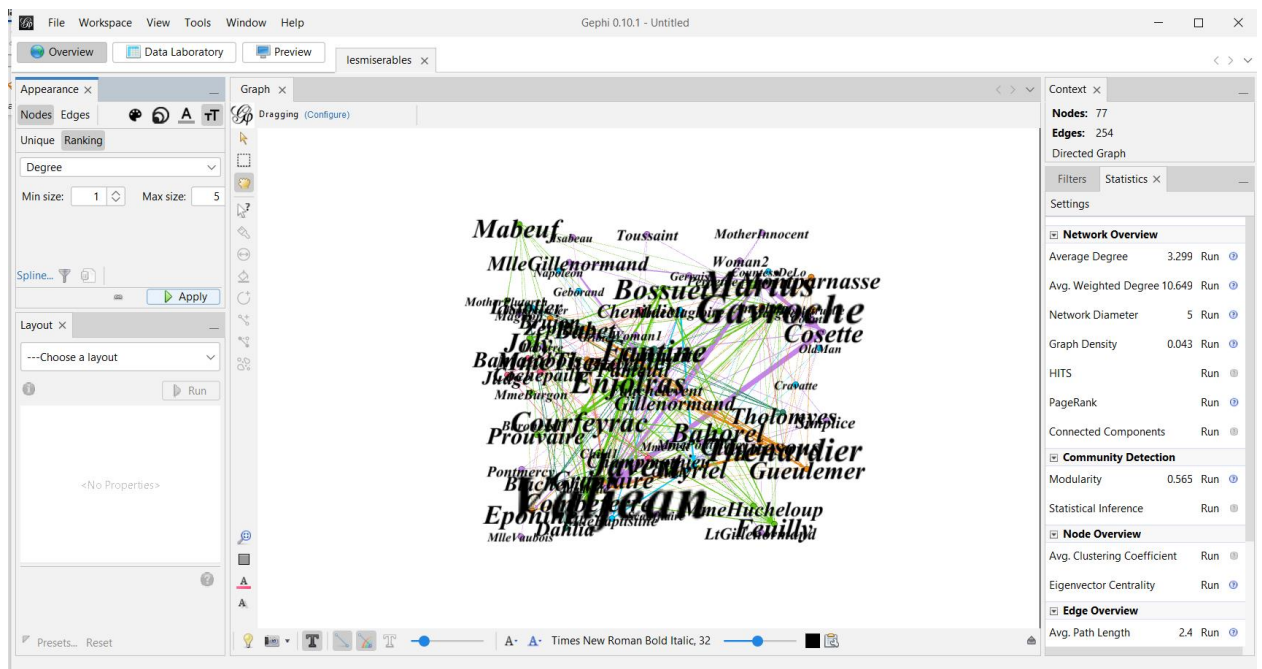
The sum change is very small (0.011672760828797354), indicating that the algorithm has converged well, and the calculated Eigenvector centrality scores are likely accurate.

The graph shows a highly skewed distribution. There are a few nodes with very high Eigenvector centrality scores (around 28), while most nodes have very low scores, clustered near zero.

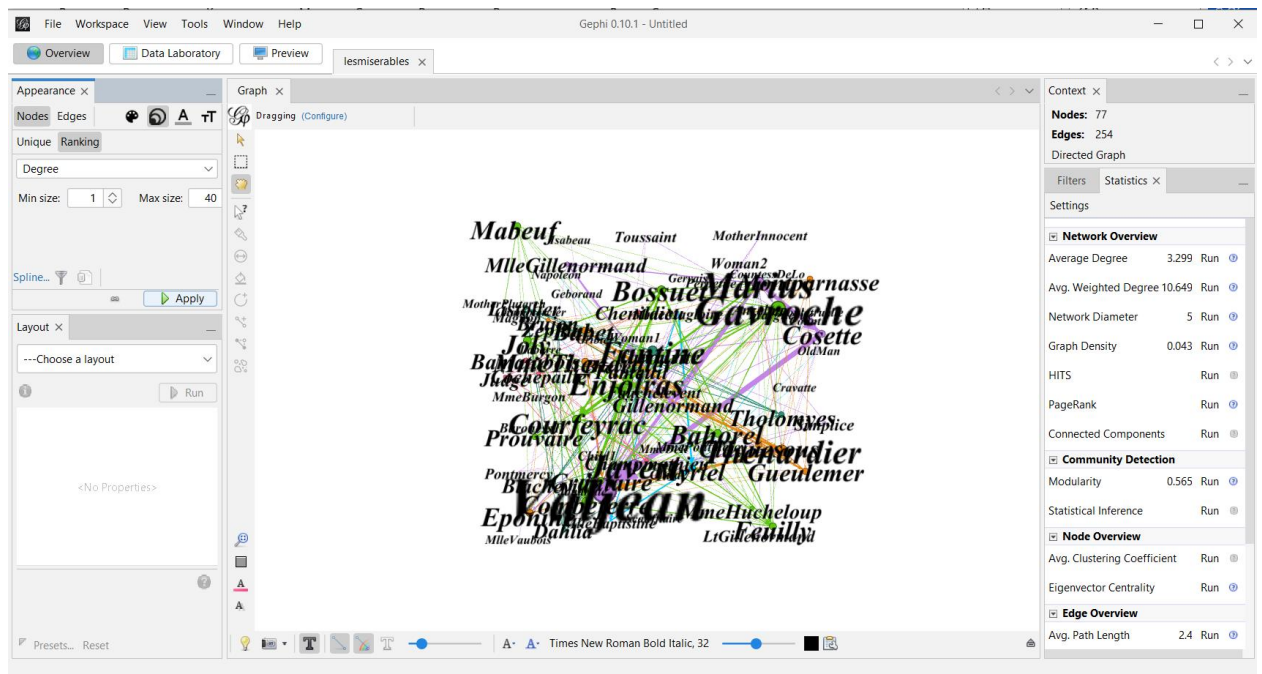
**Preview Appearance Adjustment.** Doing node color partition by using Modularity class.



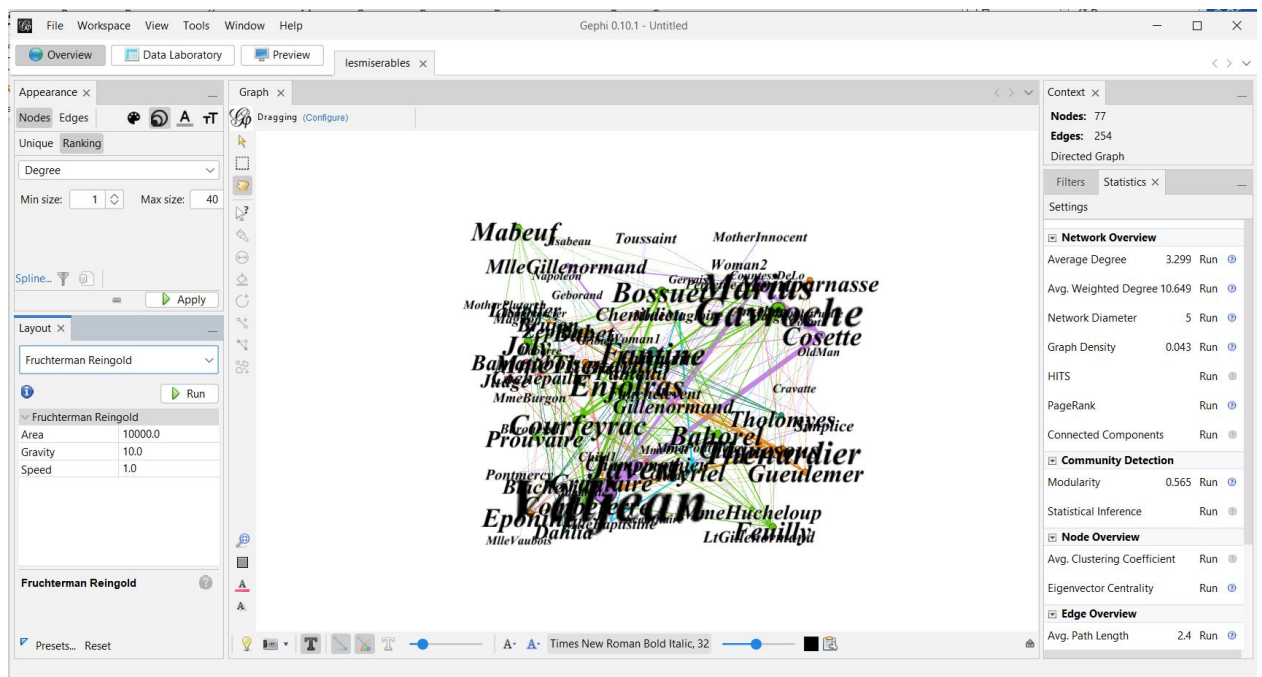
Doing label size ranking according to Degree. I changed max size to 5 because it looked better.



Doing node size ranking according to Degree. I changed max size to 40.



Doing Fruchterman Reingold layout.



Preview page with changed settings.

